



Type 2 Diabetes Mellitus

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Last updated: Jul 2026

Type 2 diabetes mellitus is characterized by insulin resistance and inadequate insulin secretion relative to needs. Early symptoms are related to hyperglycemia and include polydipsia, polyphagia, polyuria, and blurred vision. Later complications include vascular disease, peripheral neuropathy, nephropathy, and predisposition to infection. Diagnosis is by measuring plasma glucose. Treatment is diet, exercise, and medications that reduce glucose levels, including insulin, oral antihyperglycemic medications, and non-insulin injectable medications. Complications can be delayed or prevented with adequate glycemic control; cardiovascular disease remains the leading cause of mortality in diabetes mellitus.

Overview of Diabetes

VIDEO

DIABETIC KETOACIDOSIS (DKA)

Treatment

- FLUIDS for DEHYDRATION
- INSULIN to lower

Etiology of Type 2 Diabetes Mellitus

Type 2 diabetes is caused by:

- Resistance to insulin

In type 2 diabetes mellitus, insulin secretion is inadequate because patients have developed resistance to insulin (see table [General Characteristics of Types 1 and 2 Diabetes Mellitus](#)). Hepatic insulin resistance leads to an inability to suppress hepatic glucose production, and peripheral insulin resistance impairs peripheral glucose uptake. This combination gives rise to fasting and postprandial hyperglycemia. Often insulin levels are very high, especially early in the disease. Later in the course of the disease, insulin production may fall, further exacerbating hyperglycemia.

The disease generally develops in adults and becomes more common with increasing age. In the United States, approximately 30% of adults aged 18 to 44 years have impaired fasting glucose regulation or impaired glucose tolerance, while nearly 50% of adults > 65 do [\(1\)](#). In older adults, plasma glucose levels reach higher levels after eating than in younger adults, especially after meals with high carbohydrate loads. Glucose levels also take longer to return to normal, in part because of increased accumulation of visceral and abdominal fat and decreased muscle mass.

Type 2 diabetes has also become more common among children because childhood obesity has become epidemic [\(2\)](#).

More than 90% of adults with diabetes have type 2 disease [\(3, 4\)](#). There are both environmental and genetic determinants, as evidenced by the high prevalence of the disease in relatives of people with the disease [\(3, 5\)](#). Although several genetic polymorphisms have been identified, no single gene responsible for the most common forms of type 2 diabetes has been identified [\(6\)](#).

Pathogenesis is complex and incompletely understood. Hyperglycemia develops when insulin secretion can no longer compensate for insulin resistance. Although insulin resistance is characteristic in patients with type 2 diabetes and those at risk of it, patients also exhibit beta cell dysfunction and impaired insulin secretion that progresses over time, including:

- Impaired first-phase insulin secretion
- A loss of normally pulsatile insulin secretion
- An increase in proinsulin secretion signaling, indicating impaired insulin processing
- An accumulation of islet amyloid polypeptide (a protein normally secreted with insulin)

Hyperglycemia itself may impair insulin secretion, because high glucose levels desensitize beta cells, cause beta cell dysfunction (glucose toxicity), or both.

Obesity and weight gain are important determinants of insulin resistance in type 2 diabetes. They have some genetic determinants but also reflect diet, exercise, and lifestyle. An inability to suppress lipolysis in adipose tissue increases plasma levels of free fatty acids that may impair insulin-stimulated glucose transport and muscle glycogen synthase activity. Adipose tissue also functions as an endocrine organ, releasing multiple factors (adipocytokines) that favorably (adiponectin) and adversely (tumor necrosis factor-alpha, interleukin-6, leptin, resistin) influence glucose metabolism.

Intrauterine growth restriction and low birth weight have also been associated with insulin resistance in later life and may reflect adverse prenatal environmental influences on glucose metabolism ([7](#), [8](#)).

Etiology references

1. [Echouffo-Tcheugui JB, Perreault L, Ji L, Dagogo-Jack S](#). Diagnosis and Management of Prediabetes: A Review. *JAMA*. 2023;329(14):1206-1216. doi:10.1001/jama.2023.4063
2. [US Preventive Services Task Force, Mangione CM, Barry MJ, et al](#). Screening for Prediabetes and Type 2 Diabetes in Children and Adolescents: US Preventive Services Task Force Recommendation Statement. *JAMA*. 2022;328(10):963-967. doi:10.1001/jama.2022.14543
3. [American Diabetes Association Professional Practice Committee](#). 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S27-S49. doi:10.2337/dc25-S002
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Screening and Prevention of Type 2 Diabetes

Screening

Risk factors for type 2 diabetes include ([1](#)):

- Age $\geq$ 35 years
- Overweight or [obesity](#) (body mass index $> 25 \text{ kg/m}^2$ or $> 23 \text{ kg/m}^2$ in individuals of Asian descent)
- Physical inactivity
- Family history of type 2 diabetes mellitus
- History of impaired glucose regulation (prediabetes)
- [Gestational diabetes mellitus](#) or delivery of a baby $> 4.1 \text{ kg}$
- [Hypertension](#)
- [Dyslipidemia](#) (high-density lipoprotein [HDL] cholesterol $< 35 \text{ mg/dL}$ [0.9 mmol/L] or triglyceride level $> 250 \text{ mg/dL}$ [2.8 mmol/L])

- History of cardiovascular disease
- [Polyendocrine metabolic ovarian syndrome](#)
- African, Spanish, Latin American, Asian, or American Indian ancestry
- [Metabolic dysfunction-associated steatotic liver disease](#) (formerly fatty liver disease)
- [HIV infection](#)

People $\geq$ age 35 years and all adults with one or more risk factors should be screened for diabetes with a fasting plasma glucose level, glycosylated hemoglobin (HbA1C), or a 2-hour value on a 75-g oral glucose tolerance test (OGTT) at least once every 3 years as long as plasma glucose measurements are normal and at least annually if results reveal impaired fasting glucose levels (see table [Diagnostic Criteria for Diabetes Mellitus and Prediabetes](#)) (1).

Individuals taking second-generation antipsychotic medications, long-term glucocorticoids, statins, some thiazide diuretics, and some HIV medications should also be screened for prediabetes and diabetes (1).

CLINICAL CALCULATORS

[Diabetes Risk Self-Assessment](#)



CLINICAL CALCULATORS

[Body Mass Index \(Quetelet's index\)](#)



Prevention

Patients with impaired glucose regulation (prediabetes) should receive counseling addressing their risk of developing diabetes and the importance of lifestyle changes for preventing or delaying the onset of type 2 diabetes. They should be monitored closely for development of diabetes symptoms or elevated plasma glucose (via hemoglobin A1c every 6 months, and annual oral glucose tolerance testing) (2).

Type 2 diabetes usually can be prevented with lifestyle modification. Weight loss of as little as 7% of baseline body weight, combined with moderate-intensity physical activity (such as brisk walking) for at least 150 minutes/week, may reduce the incidence of diabetes in high-risk people by $> 50\%$ (2).

Various medications have been shown to prevent or slow progression of prediabetes to diabetes. These include [metformin](#), glucagon-like peptide (GLP-1) receptor agonists (eg, [liraglutide](#), [semaglutide](#), [tirzepatide](#)), [acarbose](#), thiazolidinediones, [valsartan](#), [testosterone](#), [orlistat](#), and [phentermine/topiramate](#) (2, 3). Metformin is safe and cost-effective and has the most established evidence for the prevention of diabetes. It can be considered if diet and lifestyle are not successful, especially in patients who are at highest risk for developing diabetes (body mass index ≥ 35 or history of gestational diabetes) (4, 5).

In patients with obesity, pharmacotherapy for weight loss, medical devices, or weight loss surgery can be used as adjuncts to diet and physical activity (see [weight loss in diabetes](#)). Metabolic surgery ([bariatric surgery](#)) has been shown to decrease the risk of progression to diabetes ([6](#)).

Screening and prevention references

1. [American Diabetes Association Professional Practice Committee](#). 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S27-S49. doi:10.2337/dc25-S002
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Symptoms and Signs of Type 2 Diabetes Mellitus

The mild hyperglycemia of early diabetes is often asymptomatic; therefore, diagnosis may be delayed for many years if [routine screening](#) is not performed.

Once symptomatic, the most common symptoms of diabetes mellitus are those of hyperglycemia. More significant hyperglycemia causes glycosuria and thus an osmotic diuresis, leading to urinary frequency, polyuria, and polydipsia that may progress to [orthostatic hypotension](#) and [dehydration](#). Severe dehydration causes weakness, fatigue, and mental status changes. Symptoms may come and go as plasma glucose levels fluctuate.

Polyphagia may accompany symptoms of hyperglycemia but is not typically a primary patient concern. Hyperglycemia can also cause weight loss, nausea and vomiting, and blurred vision, and it may predispose to bacterial or fungal infections.

Dermatologic Signs of Insulin Resistance



Acanthosis Nigricans

Acanthosis nigricans is skin thickening and pigmentation most typically developing in the axillae and nape of the neck (top); in people with dark skin, the skin may have a leathery appearance (bottom). It is most often a skin manifestation of impaired glucose tolerance, but it may reflect internal cancer, especially if onset is rapid and distribution is widespread.

Photos provided by Thomas Habif, MD.



Acanthosis Nigricans (Elbow Fold)

This photo shows acanthosis nigricans (hyperpigmented skin) in the elbow fold due to insulin resistance.

RICHARD USATINE MD / SCIENCE PHOTO LIBRARY



Acanthosis Nigricans (Skin Tags)

This photo shows acanthosis nigricans (hyperpigmented skin) and multiple skin tags due to insulin resistance.

In some patients, initial symptoms are those of [long-term diabetic complications](#), suggesting that the disease has been present for some time. In some patients, [hyperosmolar hyperglycemic state](#) occurs initially, especially during a period of stress or when glucose metabolism is further impaired by medications, such as glucocorticoids.

In addition to symptoms of hyperglycemia, individuals with type 2 diabetes frequently have overweight or obesity, as well as findings of insulin resistance (eg, acanthosis nigricans, skin tags).

Diagnosis of Type 2 Diabetes Mellitus

- Fasting plasma glucose, glycosylated hemoglobin (HbA1C), or oral glucose tolerance test

The diagnosis of diabetes mellitus itself is made based on an abnormal fasting plasma glucose, glycosylated hemoglobin (HbA1C), oral glucose tolerance test, or random glucose in the presence of certain symptoms. (See [Diagnosis of Diabetes Mellitus](#) and table [Diagnostic Criteria for Diabetes Mellitus and Prediabetes](#) for more details.)

Once diabetes is diagnosed, it may be classified as type 2 diabetes based on evidence of insulin resistance and relative inadequacy of insulin secretion, without evidence of autoimmune causes. The age and manner of presentation may also suggest the classification of the patient's diabetes, typically with older age at presentation than in type 1 diabetes and with associated obesity ([1](#)). Note, however, that children can and do develop type 2 diabetes, so age alone is not a reliable differentiator. (See table [General Characteristics of Types 1 and 2 Diabetes Mellitus](#) for more details.)

Prediabetes is diagnosed when measures of glucose are between the normal range and the range diagnostic for diabetes (see table [Diagnostic Criteria for Diabetes Mellitus and Prediabetes](#)). It is sometimes considered a precursor or transitional state prior to the development of type 2 diabetes. It is a significant risk factor for diabetes and may be present for many years before onset of diabetes. It is associated with an increased risk of cardiovascular disease, mortality, and [microvascular complications](#) of diabetes (retinopathy in approximately 8%, peripheral neuropathy in 7 to 16%, and chronic kidney disease in approximately 10%) ([2](#)).

Diagnosis references

1. [American Diabetes Association Professional Practice Committee](#). 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1Suppl 1):S27-S49. doi:10.2337/dc25-S002
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Treatment of Type 2 Diabetes Mellitus

- Education
- Diet and exercise
- Weight loss
- Often oral antihyperglycemics ([metformin](#), sodium glucose cotransporter 2 [SGLT2] inhibitors)
- Often non-insulin injectable medications (glucagon-like peptide-1 [GLP-1] receptor agonists and dual glucose-dependent insulinotropic polypeptide [GIP] / GLP-1 receptor agonists)
- Sometimes insulin
- To prevent complications, often renin-angiotensin-aldosterone system blockers (angiotensin-converting enzyme [ACE] inhibitors or angiotensin II receptor blockers [ARBs]), aspirin, and/or statins

Key elements of treatment for all patients are patient education, diet, exercise, weight loss, and monitoring of glucose control. Some patients with type 2 diabetes may be able to avoid or cease treatment with medications if they are able to maintain plasma glucose levels with diet and exercise alone. For detailed discussion, see [Medication Treatment of Diabetes](#).

Diabetes education

Formal diabetes education (diabetes self-management education and support or DSMES), generally conducted by diabetes nurses and nutrition specialists, is often very effective and have been shown to improve diabetes outcomes ([1](#), [2](#), [3](#), [4](#)). Education is considered as important as pharmacologic therapy in managing type 2 diabetes. Education should include information about the following:

- Causes of diabetes
- Diet
- Exercise
- Medication use, including insulin when appropriate
- Self-monitoring with fingerstick testing or continuous glucose monitoring
- Monitoring HbA1C
- Symptoms and signs of hypoglycemia, hyperglycemia, and diabetic complications

Education should be reinforced at every physician visit and hospitalization.

Diet

Individualized medical nutrition therapy with a dietitian should complement physician counseling and diabetes education; the patient and any person who prepares the patient's meals should be present ([1](#)).

Patients should be educated on consuming a diet rich in whole foods rather than processed foods. Carbohydrates should be high quality and should contain adequate amounts of fiber, vitamins, and minerals and be low in added sugar, fat, and sodium.

There are no specific recommendations on the percentages of calories that should come from carbohydrate, protein, or fat. Adjusting diet to individual circumstances can help patients control fluctuations in their glucose level. Dietary management should be individualized based on patient age, size, activity level, tastes, preferences, culture, and goals and should be formulated to accommodate requirements posed by comorbid conditions.

When appropriate, patients should be educated about using carbohydrate consumption ("counting" carbohydrates) to guide bolus insulin dosing. The concept of the glycemic index (a measure of the impact of an ingested carbohydrate-containing food on the blood glucose level) may be useful in general dietary management as well (1).

Exercise

Physical activity should increase incrementally to whatever level a patient can tolerate. Both aerobic exercise and resistance exercise have been shown to improve glycemic control in type 2 diabetes, and several studies have shown a combination of resistance and aerobic exercise to be superior to either alone (5, 6, 7). Adults with diabetes and without physical limitations should exercise for a minimum of 150 minutes/week (divided over at least 3 days) (1). Exercise has a variable short-term effect on blood glucose, depending on the timing of exercise in relation to meals and the duration, intensity, and type of exercise.

Patients with known or suspected cardiovascular disease may benefit from a pre-exercise evaluation, possibly including [exercise stress testing](#) before beginning an exercise program (1). Activity goals may require modification for patients with complications of diabetes such as [neuropathy](#) and [retinopathy](#).

Weight loss

In people with diabetes and [obesity](#), physicians should prescribe antihyperglycemic medications that promote weight loss (eg, GLP-1 receptor agonists, SGLT-2 inhibitors, or a dual GIP/GLP-1 receptor agonist) or are weight-neutral (eg, dipeptidyl peptidase-4 inhibitors, [metformin](#)) if possible (for details, see [Medication Treatment of Diabetes](#)). Specific GLP-1 receptor agonists ([semaglutide](#) and [liraglutide](#)) and a dual GIP/GLP-1 receptor agonist ([tirzepatide](#)) used for weight loss at higher doses are associated with significant weight loss even at doses used for diabetes treatment (2, 8).

Other weight loss medications, including [orlistat](#), [phentermine/topiramate](#), and [naltrexone/bupropion](#), may be useful in selected patients as part of a comprehensive weight loss program (8). [Orlistat](#), an intestinal lipase inhibitor, reduces dietary fat absorption; it reduces serum lipids and helps promote weight loss. [Phentermine/topiramate](#) is a combination medication that reduces appetite through multiple mechanisms in the brain. These medications have less weight loss efficacy than GLP-1 receptor agonists or [tirzepatide](#) but can be considered if there are contraindications or poor tolerance of more effective options.

An oral hydrogel containing cellulose and citric acid that causes patients to feel full and eat less can induce modest weight loss in patients with prediabetes and diabetes (9).

Medical weight-loss devices, including implanted gastric balloons, a vagus nerve stimulator, and gastric aspiration therapy, are also available, but their use remains limited.

[Metabolic surgery](#), such as sleeve gastrectomy or gastric bypass, also leads to weight loss and improvement in glucose control (independent of weight loss) and decreased cardiovascular risk in patients who have diabetes mellitus and should be recommended for appropriately selected patients (8). Consideration of metabolic surgery is specifically recommended in patients with diabetes and a body mass index ≥ 30 kg/m². Rapid weight loss induced by GLP-1 receptor agonist pharmacotherapy or bariatric surgery can lead to malnutrition (undernutrition) or sarcopenia. A diet with sufficient protein and resistance exercise should be encouraged, and patients should be tested for micronutrient deficiencies.

Pharmacotherapy

(See also [Medications for Diabetes Mellitus Treatment](#).)

Treatment goals in type 2 diabetes include glycemic control, weight management, and risk reduction in patients with comorbidities such as [atherosclerotic cardiovascular disease](#) (ASCVD), [heart failure](#), [chronic kidney disease](#) (CKD), and [metabolic dysfunction-associated steatohepatitis](#). [Monitoring](#) treatment effectiveness is discussed elsewhere.

(See [Weight Loss](#) for a discussion of relevant pharmacotherapy.)

Initial medication selection

Patients with type 2 diabetes and mildly elevated plasma glucose (HbA1C approximately 7.5%) may be prescribed a trial of diet and exercise. For patients with an inadequate response to a trial of diet and exercise, or in those with more significant hyperglycemia, pharmacologic therapy is necessary.

In patients without ASCVD, heart failure, or CKD, selection of therapy involves consideration of adverse effects, convenience, cost, and patient preference. [Metformin](#) is usually the first oral medication used due to its cost-effectiveness and safety profile. GLP-1 or dual GIP/GLP-1 receptor agonists are an effective second-line therapy after metformin and may be more effective than insulin; they may also be used as an add-on to insulin (10, 11). Patients with obesity can also benefit from the weight-lowering effects of GLP-1 receptor agonist therapy or from the use of [tirzepatide](#), a dual GIP/GLP-1 receptor agonist (12).

In patients with more significant glucose elevations at diagnosis or with HbA1C levels 1.5 to 2.0% above target, early combination therapy ([metformin](#) plus a dipeptyl-peptidase-4 inhibitor, a GLP-1 or dual GIP/GLP-1 receptor agonist, or a SGLT2 inhibitor), and/or insulin therapy should be initiated. There is evidence that early combination medication therapy with vildagliptin and [metformin](#) leads to superior and more durable blood glucose control than a stepwise approach to adding diabetes pharmacotherapy (10, 13).

Medications in patients with comorbidities

In patients with known **ASCVD** or at moderate or high risk for ASCVD (> 3 cardiovascular risk factors not including diabetes), or with known CKD, a SGLT2 inhibitor or a GLP-1 receptor agonist is recommended because these medication classes decrease major adverse cardiovascular events (eg, myocardial infarction, stroke) and mortality ([10](#), [12](#)). Specific cardiovascular risk factors used for this determination include:

- Poor glycemic control
- Hypertension
- Dyslipidemia
- Smoking or tobacco use
- Harmful alcohol use
- Family history of ASCVD or CKD
- Obesity

For patients with type 2 diabetes, a statin is recommended for all adults 40 to 75 years old, for those 20 to 39 years old with additional risk factors for ASCVD, and those of any age with CKD ([14](#), [15](#), [16](#)). The intensity of statin therapy depends upon specific risk factors. The risk of teratogenicity should be considered in all individuals of childbearing potential.

Aspirin is recommended for patients with diabetes and known ASCVD ([14](#)).

Patients with type 2 diabetes and **heart failure** with either reduced or preserved ejection fraction should be treated with an SGLT2 inhibitor. Those with symptomatic heart failure with preserved ejection fraction and obesity may also receive a GLP-1 receptor agonist ([10](#)).

In patients with **CKD** ([17](#), [18](#)) or heart failure ([19](#)) without contraindications, SGLT2-inhibitors are recommended because they have been shown to decrease disease progression and mortality. An ACE inhibitor or ARB is recommended for patients with diabetes and CKD (evidenced by albuminuria and/or decreased glomerular filtration rate), and is first-line therapy for patients with diabetes and hypertension regardless of CKD status ([15](#)). A nonsteroidal mineralocorticoid receptor antagonists (eg, **finerenone**) is recommended in individuals with CKD, particularly those with persistent albuminuria despite maximal ACE inhibitor or ARB therapy ([12](#)).

GLP-1 or dual GIP/GLP-1 receptor agonists and **pioglitazone** are recommended for patients with **metabolic dysfunction-associated steatotic liver disease** or **metabolic dysfunction-associated steatohepatitis** ([10](#)).

Insulin

Insulin should generally be considered in patients with evidence of ongoing catabolism (unintentional weight loss) or symptoms of hyperglycemia (ie, polyuria, polydipsia) with HbA1C levels > 10% (86 mmol/mol) and blood glucose levels ≥ 300 mg/dL (16.7 mmol/L) ([10](#)).

Insulin is also indicated as initial therapy for patients with type 2 diabetes who are pregnant ([20](#)), and for patients who present with acute metabolic decompensation, such as **hyperosmolar hyperglycemic state**

or [diabetic ketoacidosis](#) (DKA). Patients with severe hyperglycemia may respond better to therapy after glucose levels are normalized with insulin treatment.

Depending on patient needs, [insulin](#) can be administered as basal insulin (long- or intermediate-acting insulin), bolus insulin with meals (rapid acting insulin), a combination of basal and bolus insulin, or as twice daily premixed intermediate- and short- or rapid-acting insulin.

Foot care

Adults with diabetes, including type 2 diabetes, should undergo a comprehensive foot examination at least annually (see [Complications](#) for more details) ([21](#), [22](#), [23](#)). Individuals with peripheral neuropathy, history of ulcers or amputations, or poorly controlled glucose, should have an examination at every visit and certain guidelines, recommend this approach for all patients with diabetes. Assessment of neuropathy includes 10 g monofilament test as well as pinprick, temperature, vibration testing.

Regular professional podiatric care, including trimming of toenails and calluses, is important for patients with sensory loss or circulatory impairment. Toenails should be cut, preferably by a podiatrist, straight across and not too close to the skin.

Patients with sensory loss or circulatory impairment should be advised to inspect their feet daily for cracks, fissures, calluses, corns, and ulcers. Feet should be washed daily in lukewarm water, using mild soap, and dried gently and thoroughly. A lubricant (eg, [lanolin](#)) should be applied to dry, scaly skin. Nonmedicated foot powders should be applied to moist feet. Adhesive plasters and tape, harsh chemicals, corn cures, water bottles, and electric pads should not be used on skin. Patients should change socks or stockings daily and not wear constricting clothing (eg, garters, socks, stockings with tight elastic tops).

Shoes should fit well, be wide-toed without open heels or toes, and be changed frequently. Special shoes should be prescribed to reduce trauma if the foot is deformed (eg, previous toe amputation, [hammer toe](#), [bunion](#)). Walking barefoot should be avoided.

Patients with [neuropathic foot ulcers](#) ideally should avoid weight bearing until ulcers heal. If they cannot, they should wear appropriate orthotic protection. Debridement and antibiotics may result in good healing and may prevent major surgery. After the ulcer has healed, appropriate inserts or special shoes should be prescribed. In refractory cases, especially if [osteomyelitis](#) is present, surgical removal of the metatarsal head (the source of pressure), amputation of the involved toe, or transmetatarsal amputation may be required. A neuropathic joint can often be satisfactorily managed with orthopedic devices (eg, short leg braces, molded shoes, sponge-rubber arch supports, crutches, prostheses).

Vaccination

All patients with diabetes, should be vaccinated against [Streptococcus pneumoniae](#), [influenza virus](#), [hepatitis B](#), [varicella](#), [respiratory syncytial virus](#), and [SARS-CoV-2](#) as per standard recommendations ([24](#)).

Treatment references

Monitoring Type 2 Diabetes Treatment

HbA1C levels reflect glucose control over the preceding 3 months and hence assess control between physician visits. HbA1C should be assessed at least twice a year in patients with type 2 diabetes when plasma glucose appears stable and more frequently when control is uncertain. For most patients, the goal HbA1C is < 7%; however, this goal should be individualized. For older adults, those with high risk of hypoglycemia, and those with limited life expectancy, a target HbA1C of < 7.5 to 8.5% may be appropriate ([1](#), [2](#)). Home HbA1c testing kits are available but are used infrequently.

The frequency of home blood glucose monitoring or need for continuous glucose monitoring in type 2 diabetes should be tailored to the individual depending upon the risk of hypoglycemia, the use of insulin, and overall glycemic control (HbA1c level). Patients treated with insulin should monitor glucose at least once daily, and patients taking multiple daily injections of insulin should use continuous glucose monitoring when possible, or check fingerstick glucose multiple times per day, similar to [monitoring in a patient with type 1 diabetes](#). Even for patients whose type 2 diabetes is relatively well-controlled without insulin, continuous glucose monitoring can help support behavioral modification to understand the impact of dietary choices and exercise on glucose levels.

Monitoring references

1. [American Diabetes Association Professional Practice Committee](#). 6. Glycemic Goals and Hypoglycemia: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S128-S145. doi:10.2337/dc25-S006
2. [Kalyani RR, Neumiller JJ, Maruthur NM, Wexler DJ](#). Diagnosis and Treatment of Type 2 Diabetes in Adults: A Review. *JAMA*. Published online June 23, 2025. doi:10.1001/jama.2025.5956

Complications of Type 2 Diabetes

Most complications that occur in patients with type 2 diabetes are the vascular, neurologic, and immunologic complications of diabetes in general; metabolic-dysfunction associated liver disease is more specific to type 2 diabetes. Screening recommendations exist for both general and specific complications.

Acute complications of type 2 diabetes and its treatment, including [hyperosmolar hyperglycemic state](#) and [hypoglycemia](#), are discussed elsewhere.

See [Long-Term Complications of Diabetes Mellitus](#) for a more detailed discussion of specific complications.

Complications of diabetes in general

Chronic hyperglycemia leads to complications that affect small vessels (microvascular), large vessels (macrovascular), or both.

Microvascular complications include [retinopathy](#), [nephropathy](#), and [neuropathy](#). Neuropathy is a heterogeneous long-term complication, with multifactorial pathogenesis that includes toxic effects of hyperglycemia and advanced glycation end-products on nerves, and ischemia to nerves from microvascular disease.

Macrovascular complications include [atherosclerosis](#) of large vessels leading to [angina pectoris](#) and [myocardial infarction](#), [transient ischemic attacks](#) and [strokes](#), and [peripheral artery disease](#).

Immune dysfunction is another major long-term complication.

Risk of [complications](#) can be decreased by strict glycemic control and by management of [hypertension](#) and [lipid levels](#). Specific measures for prevention of progression of complications once detected are described under [Long-Term Complications](#).

Complications specific to type 2 diabetes

The majority, but not all, adults with type 2 diabetes also have [metabolic dysfunction-associated steatotic liver disease](#) (MASLD), placing them at risk for the development of metabolic dysfunction-associated steatohepatitis (MASH) and cirrhosis ([1](#), [2](#)).

Screening for complications in type 2 diabetes

Adults with type 2 diabetes should undergo the following screening for complications ([1](#), [3](#), [4](#), [5](#)):

- Retinopathy: Dilated and comprehensive eye examination at diagnosis, and then every 1 to 2 years
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- Neuropathy and foot ulcers: Foot examination with pulses, reflexes, temperature or pinprick sensation, vibration, and monofilament testing at diagnosis, then annually
- Nephropathy: Spot urine albumin-to-creatinine ratio and estimated glomerular filtration rate (eGFR) at diagnosis, then annually
- Hypertension: Blood pressure measurement at every visit
- Atherosclerotic cardiovascular disease: Lipid profile soon after diagnosis, then annually; additional screening depends on signs, symptoms, and additional risk factors
- Heart failure: Consider measuring a natriuretic peptide
- Peripheral artery disease: Consider ankle-brachial index if ≥ 65 years and any other microvascular disease, foot complications, or end-organ damage is present
- MASH and cirrhosis: Calculate the fibrosis-4 (FIB-4) index using alanine aminotransferase (ALT), aspartate aminotransferase (AST), and platelet count

Additional cardiovascular screening may include risk stratification, electrocardiography, or other testing.

Complications references

CLINICAL CALCULATORS

[Cardiovascular Risk Assessment \(10-year, Revised Pooled Cohort Equations 2018\)](#)



1. [American Diabetes Association Professional Practice Committee](#). 4. Comprehensive Medical Evaluation and Assessment of Comorbidities: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S59-S85. doi:10.2337/dc25-S004
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Key Points

- Type 2 diabetes is caused by hepatic insulin resistance (causing an inability to suppress hepatic glucose production) and peripheral insulin resistance (which impairs peripheral glucose uptake), in combination with a pancreatic beta cell secretory defect.
- Diagnose by elevated fasting plasma glucose level and/or elevated hemoglobin A1C and/or elevated 2-hour value on oral glucose tolerance test.
- Do regular screening for long-term complications.
- Microvascular complications include nephropathy, neuropathy, and retinopathy.
- Macrovascular complications involve atherosclerosis resulting in coronary artery disease, transient ischemic attack/stroke, and peripheral arterial insufficiency.
- Treat with diet, exercise, weight loss, oral or injectable antihyperglycemic medications, and sometimes insulin.
- Often, give renin-angiotensin-aldosterone system blockers, and statins to prevent complications.

More Information

The following English-language resources may be useful. Please note that The Manual is not responsible for the content of these resources.

[American Diabetes Association Professional Practice Committee](#). Standards of Care in Diabetes-2025. *Diabetes Care* 2025;48(1 Suppl 1):S1-S5. doi:10.2337/dc25-SINT

[Endocrine Society: Clinical Practice Guidelines](#): provides guidelines on evaluation and management of patients with diabetes as well as links to other information for clinicians

Drug Information for the Topic



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Acute Complications of Diabetes Mellitus

Full Review: Dec 2025 By [Erika F. Brutsaert](#), MD, New York Medical College | Peer reviewed by [Glenn D. Braunstein](#), MD, Cedars-Sinai Medical Center
Last updated: May 2026

Acute complications of diabetes include diabetic ketoacidosis and hyperosmolar hyperglycemic state. Alcoholic ketoacidosis, though not a complication of diabetes, is discussed here due to overlap in physiology and treatment.

Diabetic Ketoacidosis

Diabetic ketoacidosis (DKA) is an acute metabolic complication of diabetes characterized by hyperglycemia, hyperketonemia, and metabolic acidosis. Hyperglycemia causes an osmotic diuresis with significant fluid and electrolyte loss. DKA occurs mostly in type 1 diabetes mellitus. It causes nausea, vomiting, and abdominal pain and can progress to cerebral edema, coma, and death. DKA is diagnosed by detection of hyperketonemia and anion gap metabolic acidosis in the presence of hyperglycemia. Treatment involves volume expansion, insulin replacement, and prevention of hypokalemia.

Diabetic ketoacidosis (DKA) occurs primarily in patients with type 1 diabetes mellitus and is less common in those with type 2 diabetes. It develops when insulin levels are insufficient to meet the body's basic metabolic requirements. DKA is the first manifestation of type 1 diabetes in approximately one-third of patients ([1](#)). Insulin deficiency can be absolute (eg, during lapses in the administration of exogenous insulin) or relative (eg, when usual insulin doses do not meet metabolic needs during physiologic stress).

Common **physiologic stresses** that can trigger DKA include:

- Acute infection (eg, [pneumonia](#), [urinary tract infection](#), [COVID-19](#))
- [Myocardial infarction](#)
- [Stroke](#)
- [Pancreatitis](#)
- Pregnancy
- Trauma
- Missed insulin doses

Some **medications** implicated in causing DKA include:

- Sodium glucose cotransporter 2 (SGLT-2) inhibitors
- Glucocorticoids
- Thiazide diuretics
- Sympathomimetics

DKA is less common in type 2 diabetes mellitus, but it may occur in situations of unusual physiologic stress. Ketosis-prone type 2 diabetes (also referred to as Flatbush diabetes) is a variant of type 2 diabetes, which sometimes occurs in patients with obesity, often those with African (including African American or Afro-Caribbean) ancestry (2). Patients with ketosis-prone diabetes can have significant impairment of beta-cell function with hyperglycemia, and are therefore more likely to develop DKA when significant hyperglycemia occurs.

SGLT-2 inhibitors have been implicated in causing DKA in both type 1 and type 2 diabetes (3). In pregnant patients and in patients taking SGLT2 inhibitors, DKA may occur at lower or even normal blood glucose levels (euglycemic DKA). Euglycemic DKA (ketoacidosis without elevated glucose levels) can also occur with alcohol overuse or cirrhosis.

General references

1. [Dovc K, Neuman V, Gita G, et al.](#) Association of Diabetic Ketoacidosis at Onset, Diabetes Technology Uptake, and Clinical Outcomes After 1 and 2 Years of Follow-up: A Collaborative Analysis of Pediatric Registries Involving 9,269 Children With Type 1 Diabetes From Nine Countries. *Diabetes Care*. 2025;48(4):648-654. doi:10.2337/dc24-2483
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3. [Wang Y, Qin Y, Zhang J, et al.](#) Sodium-Glucose Cotransporter-2 Inhibitors and Diabetic-Ketoacidosis in T2DM Patients: An Updated Meta-Analysis and a Mendelian Randomization Analysis. *Clin Pharmacol Ther*. 2025;117(6):1661-1669. doi:10.1002/cpt.3615

Pathophysiology of DKA

Insulin deficiency and an increase in counterregulatory hormones (glucagon, catecholamines, cortisol) causes the body to metabolize triglycerides and amino acids instead of glucose for energy. Serum levels of glycerol and free fatty acids rise because of unrestrained lipolysis. Alanine levels rise because of muscle catabolism. Glycerol and alanine provide substrate for hepatic gluconeogenesis, which is stimulated by the excess of glucagon that accompanies insulin deficiency.

Glucagon also stimulates mitochondrial conversion of free fatty acids into ketones. Insulin normally blocks ketogenesis by inhibiting the transport of free fatty acid derivatives into the mitochondrial matrix, but ketogenesis proceeds in the absence of insulin. The major ketoacids produced, acetoacetic acid and beta-hydroxybutyric acid, are strong organic acids that create [metabolic acidosis](#). Acetone

derived from the metabolism of acetoacetic acid accumulates in serum and is slowly disposed of by respiration.

Hyperglycemia due to insulin deficiency causes an osmotic diuresis that leads to marked urinary losses of water and electrolytes. Urinary excretion of ketones obligates additional losses of sodium and potassium. Serum sodium may fall due to natriuresis or rise due to excretion of large volumes of free water.

Potassium is also lost in large quantities. Despite a significant total body deficit of potassium, initial serum potassium is typically normal or elevated because of the extracellular migration of potassium in response to acidosis. Potassium levels generally fall further during treatment as insulin therapy drives potassium into cells. If serum potassium is not monitored and replaced as needed, life-threatening [hypokalemia](#) may develop.

Symptoms and Signs of DKA

Symptoms and signs of diabetic ketoacidosis include [symptoms of hyperglycemia](#) with the addition of nausea, vomiting, and—particularly in children—abdominal pain. Lethargy and somnolence are symptoms of more severe decompensation. Patients may be hypotensive and tachycardic due to dehydration and acidosis; they may breathe rapidly and deeply to compensate for acidemia (Kussmaul respirations). They may also have fruity breath due to exhaled acetone. Fever is not a sign of DKA itself and, if present, signifies underlying infection. In the absence of timely treatment, DKA progresses to coma and death.

Acute cerebral edema, a complication in less than 1% of patients with DKA ([1](#), [2](#)), occurs primarily in children and less often in adolescents and young adults. Headache and fluctuating level of consciousness herald this complication in some patients, but respiratory arrest is the initial manifestation in others. The cause is not well understood but may be related to too-rapid reductions in serum osmolality or to brain ischemia. In studies primarily of children, those the highest blood urea nitrogen (BUN) levels and lowest PaCO₂ at presentation appear to be at greatest risk. Delays in correction of hyponatremia, and the use of bicarbonate during DKA treatment are other possible risk factors ([3](#)).

Symptoms and signs references

1. [Edge JA, Hawkins MM, Winter DL, Dunger DB](#). The risk and outcome of cerebral oedema developing during diabetic ketoacidosis. *Arch Dis Child*. 2001;85(1):16-22. doi:10.1136/adc.85.1.16
2. [Lawrence SE, Cummings EA, Gaboury I, Daneman D](#). Population-based study of incidence and risk factors for cerebral edema in pediatric diabetic ketoacidosis. *J Pediatr*. 2005;146(5):688-692. doi:10.1016/j.jpeds.2004.12.041
3. [Nigrovic LE, Kuppermann N, Ghetti S, et al](#). Emergency Department Presentations of Diabetic Ketoacidosis in a Large Cohort of Children. *Pediatr Diabetes*. 2023;2023:6693226. doi:10.1155/2023/6693226

Diagnosis of DKA

- Arterial pH
- Serum glucose
- Serum and urine ketones
- Calculation of anion gap
- Serum electrolytes, BUN, and creatinine

In patients suspected of having diabetic ketoacidosis, serum electrolytes, blood urea nitrogen (BUN) and creatinine, glucose, ketones, and osmolality should be measured. Urine should be tested for ketones. Patients who appear significantly ill and those with positive ketones should have arterial blood gas measurement.

Diagnostic criteria for DKA include ([1](#), [2](#), [3](#)):

- Arterial or venous pH < 7.30
- Anion gap > 10 to 12 mEq/L (10 to 12 mmol/L)
- Serum bicarbonate < 15 to 18 mEq/L (15 to 18 mmol/L)
- Presence of serum or urine ketones
- Serum glucose > 200 to 250 mg/dL (11.1 to 13.8 mmol/L)

Some guidelines do not require a specific serum glucose level for the diagnosis of DKA and allow for the presence of known diabetes to substitute for a specific glucose cutoff due to the recognition that DKA can occur in patients with normal or mildly elevated glucose levels (euglycemic DKA) ([1](#), [2](#), [3](#)).

A presumptive diagnosis may be made when urine glucose and ketones are positive on urinalysis. Urine test strips and some assays for serum ketones may underestimate the degree of ketosis because they detect acetoacetic acid and not beta-hydroxybutyric acid, which is usually the predominant ketoacid. Blood beta-hydroxybutyrate can be measured, or treatment can be initiated based on clinical suspicion and the presence of anion gap acidosis if serum or urine ketones are low.

Symptoms and signs of a triggering illness should be pursued with appropriate studies (eg, cultures, imaging studies). Adults should have an ECG to screen for acute myocardial infarction and to help determine the significance of abnormalities in serum potassium.

Other laboratory abnormalities include:

- [Hyponatremia](#)
- Elevated serum creatinine
- Elevated plasma osmolality
- Hypokalemia, after correction of acidosis
- Elevated serum amylase and lipase

Hyperglycemia may cause dilutional hyponatremia, so measured serum sodium is corrected by adding 1.6 mEq/L (1.6 mmol/L) for each 100 mg/dL (5.6 mmol/L) elevation of serum glucose over 100 mg/dL (5.6 mmol/L).

CLINICAL CALCULATORS

[Sodium Correction in Hyperglycemia \(Katz 1973\)](#)



As acidosis is corrected, serum potassium drops. An initial potassium level < 4.5 mEq/L (< 4.5 mmol/L) indicates marked potassium depletion and requires immediate potassium supplementation.

Serum amylase and lipase are often elevated, even in the absence of [pancreatitis](#) (which may be present in patients with [alcoholic ketoacidosis](#) and in those with coexisting hypertriglyceridemia).

CLINICAL CALCULATORS

[Anion Gap](#)



Diagnosis references

1. [Buse JB, Wexler DJ, Tsapas A, et al](#): 2019 Update to: Management of Hyperglycemia in Type 2 Diabetes, 2018. A Consensus Report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD). *Diabetes Care* 43(2):487–493, 2020. doi: 10.2337/dci19-0066
2. [Garber AJ, Handelsman Y, Grunberger G, et al](#): Consensus statement by the American Association of Clinical Endocrinologists and American College of Endocrinology on the comprehensive type 2 diabetes management algorithm--2020 executive summary. *Endocrine Practice* 26:107–139, 2020.

Treatment of DKA

- Fluid resuscitation with IV 0.9% saline
- Correction of hypokalemia
- IV insulin (as long as serum potassium is ≥ 3.3 mEq/L [3.3 mmol/L])
- Rarely IV [sodium bicarbonate](#) (if pH < 7 after 1 hour of treatment)

The most urgent goals for treating diabetic ketoacidosis are rapid intravascular volume repletion, correction of hyperglycemia and acidosis, and prevention of hypokalemia ([1](#), [2](#), [3](#)). Identification of precipitating factors is also important.

Treatment of most cases of DKA should occur in intensive care settings because clinical and laboratory assessments are initially needed every hour or every other hour with appropriate adjustments in treatment. Milder cases may be able to be managed with subcutaneous [insulin](#) every 1 to 2 hours, in a step-down or emergency room.

Volume repletion

Intravascular volume should be restored rapidly to raise blood pressure and ensure glomerular perfusion ([1](#), [2](#), [3](#)). Initial volume resuscitation in adults is typically achieved with rapid IV infusion of 1 to 1.5 L of 0.9% saline solution in the first hour, followed by saline infusions at 250 to 500 mL/hour. Additional boluses or a faster rate of infusion may be needed to correct hypotension. Slower rates of infusion may be needed in patients with heart failure or in those at risk for volume overload. If the serum sodium level is normal or high, the normal saline is replaced by 0.45% saline after initial volume resuscitation. Once intravascular volume is restored, remaining total body water deficits are corrected more slowly, typically over about 24 hours. When plasma glucose falls to < 200 mg/dL (< 11.1 mmol/L), IV fluid should be changed, and 5% to 10% [dextrose](#) can be added to 0.45% saline.

Correction of hyperglycemia and acidosis

In adults, an initial bolus of [regular insulin](#) 0.1 unit/kg IV bolus may be given. Then, whether or not a bolus has been given, adults should be administered a continuous IV infusion of regular insulin 0.1 unit/kg/hour in 0.9% saline solution. *[Insulin](#) should be withheld until serum potassium is ≥ 3.3 mEq/L (≥ 3.3 mmol/L).* Insulin adsorption onto IV tubing can lead to inconsistent effects, which can be minimized by preflushing the IV tubing with insulin solution. If plasma glucose does not fall by 50 to 75 mg/dL (2.8 to 4.2 mmol/L) in the first hour, insulin doses should be doubled.

Ketones should begin to clear within hours if insulin is given in sufficient doses. However, clearance of ketones may appear to lag because of conversion of beta-hydroxybutyrate to acetoacetate (the ketone measured in most hospital laboratories) as acidosis resolves.

Serum pH and bicarbonate levels should also quickly improve, but restoration of a normal serum bicarbonate level may take 24 hours. Bicarbonate should not be given routinely because it can lead to development of acute cerebral edema (primarily in children). If bicarbonate is used, it should be started only if the pH is < 7 ([4](#)), and only modest pH elevation should be attempted with doses of 50 to 100 mEq (50 to 100 mmol) given over 2 hours, followed by repeat measurement of arterial pH and serum potassium.

When plasma glucose becomes < 250 mg/dL (< 13.9 mmol/L) in adults, 5% to 10% [dextrose](#) should be added to IV fluids to reduce the risk of hypoglycemia ([2](#), [4](#)). The dextrose concentration can be adjusted and the insulin dose can be reduced to maintain glucose 150 to 200 mg/dL (8.3 to 11.1 mmol/L), but the continuous IV infusion of regular insulin should be maintained until the anion gap has narrowed on 2 consecutive blood tests and blood and urine are consistently negative for ketones. A longer duration of treatment with insulin and dextrose may be required in DKA associated with SGLT-2 inhibitor use.

When the patient is stable and able to eat, a typical [basal-bolus insulin regimen](#) is begun. IV insulin should be continued for 2 hours after the initial dose of basal subcutaneous insulin is given.

Hypokalemia prevention

Prevention of [hypokalemia](#) requires replacement of 20 to 30 mEq (20 to 30 mmol) potassium in each liter of IV fluid to keep serum potassium between 4 and 5 mEq/L (4 and 5 mmol/L). If serum potassium is < 3.3 mEq/L (< 3.3 mmol/L), insulin should be withheld (or stopped temporarily) and potassium given at 40 mEq/hour until serum potassium is ≥ 3.3 mEq/L (≥ 3.3 mmol/L); if serum potassium is > 5 mEq/L (> 5 mmol/L), potassium supplementation can be withheld.

Initially normal or elevated serum potassium measurements may reflect shifts from intracellular stores in response to acidemia and belie the true potassium deficits that almost all patients with DKA have. Insulin replacement rapidly shifts potassium into cells, so levels should be checked hourly or every other hour in the initial stages of treatment.

Other measures

[Hypophosphatemia](#) often develops during treatment of DKA, but phosphate repletion does not appear to be of benefit except for cases of severe hypophosphatemia (< 0.78 mg/dL [<0.32 mmol/L]) (5). If indicated (eg, if rhabdomyolysis, hemolysis, or neurologic deterioration occurs), [potassium phosphate](#), 1 to 2 mmol/kg of phosphate, can be infused over 6 to 12 hours. If [potassium phosphate](#) is given, the serum calcium level usually decreases and should be monitored.

Treatment of suspected cerebral edema is hyperventilation, glucocorticoids, and [mannitol](#), but these measures are often ineffective after the onset of respiratory arrest.

Treatment references

1. [American Diabetes Association Professional Practice Committee](#). 6. Glycemic Goals and Hypoglycemia: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S128-S145. doi:10.2337/dc25-S006
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3. [Wolfsdorf JJ, Glaser N, Agus M, et al](#). ISPAD Clinical Practice Consensus Guidelines 2018: Diabetic ketoacidosis and the hyperglycemic hyperosmolar state. *Pediatr Diabetes*. 2018;19 Suppl 27:155-177. doi:10.1111/pedi.12701
4. [American Diabetes Association Professional Practice Committee](#). 16. Diabetes Care in the Hospital: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S321-S334. doi:10.2337/dc25-S016
5. [Tran TTT, Pease A, Wood AJ, et al](#). Review of Evidence for Adult Diabetic Ketoacidosis Management Protocols. *Front Endocrinol (Lausanne)*. 2017;8:106. doi:10.3389/fendo.2017.00106

Prognosis for DKA

Overall mortality rates for diabetic ketoacidosis are < 1% (1); however, mortality is higher in older patients and in patients with other life-threatening illnesses. Type 2 (versus type 1) diabetes, sepsis,

altered level of consciousness, and obesity are also associated with an increased risk of mortality ([2](#), [3](#)). Cerebral edema in adults with DKA is thought to be very rare.

Prognosis references

1. [Umpierrez G, Korytkowski M](#). Diabetic emergencies - ketoacidosis, hyperglycaemic hyperosmolar state and hypoglycaemia. *Nat Rev Endocrinol*. 2016;12(4):222-232. doi:10.1038/nrendo.2016.15
2. [Eledrisi MS, Alkabbani H, Aboawon M, et al](#). Clinical characteristics and outcomes of care in patients hospitalized with diabetic ketoacidosis. *Diabetes Res Clin Pract*. 2022;192:110041. doi:10.1016/j.diabres.2022.110041
3. [Sato Y, Morita K, Okada A, Matsui H, Fushimi K, Yasunaga H](#). Factors affecting in-hospital mortality of diabetic ketoacidosis patients: A retrospective cohort study. *Diabetes Res Clin Pract*. 2021;171:108588. doi:10.1016/j.diabres.2020.108588

Key Points

- Diabetic ketoacidosis (DKA) is an acute metabolic complication of diabetes characterized by hyperglycemia, hyperketonemia, and metabolic acidosis.
- DKA can occur when acute physiologic stressors (eg, infections, myocardial infarction) trigger acidosis, moderate glucose elevation, dehydration, and severe potassium loss, particularly in patients with type 1 diabetes.
- Diagnose by an arterial pH < 7.30, the presence of serum or urine ketones, hyperglycemia, decreased serum bicarbonate and/or increased anion gap.
- Correct acidosis and hyperglycemia with IV fluid and insulin.
- Withhold insulin until serum potassium is ≥ 3.3 mEq/L (≥ 3.3 mmol/L).
- Acute cerebral edema is a rare but lethal complication.

Hyperosmolar Hyperglycemic State (HHS)

Hyperosmolar hyperglycemic state is a metabolic complication of diabetes mellitus characterized by severe hyperglycemia, extreme dehydration, hyperosmolar plasma, and altered consciousness. It most often occurs in type 2 diabetes, often in the setting of physiologic stress. Hyperosmolar hyperglycemic state is diagnosed by severe hyperglycemia and plasma hyperosmolality in the absence of significant ketosis. Treatment is IV saline solution and insulin. Complications include coma, seizures, and death.

Hyperosmolar hyperglycemic state (previously referred to as hyperglycemic hyperosmolar nonketotic coma [HHNK] and nonketotic hyperosmolar syndrome [NKHS]) is a complication of type 2 diabetes mellitus and has an estimated mortality rate of up to approximately 2%, higher than the mortality for [diabetic ketoacidosis](#) (currently < 1%) (1).

It usually develops after a period of symptomatic hyperglycemia in which fluid intake is inadequate to prevent extreme dehydration due to the hyperglycemia-induced osmotic diuresis.

Precipitating factors include:

- Acute infections and other medical conditions
- Medications that impair glucose tolerance (glucocorticoids) or increase fluid loss (diuretics)
- Nonadherence to diabetes treatment

Serum ketones are not present because the amount of insulin present in most patients with type 2 diabetes is adequate to suppress ketogenesis. Because symptoms of acidosis are not present, most patients endure a significantly longer period of osmotic diuresis (high solute concentrations from glucose in the renal tubules, leading to excess water loss). This causes more severe dehydration before presentation, and thus plasma glucose (> 600 mg/dL [> 33.3 mmol/L]) and osmolality (> 320 mOsm/L) are typically much higher than in diabetic ketoacidosis.

General reference

1. [American Diabetes Association Professional Practice Committee](#). 6. Glycemic Goals and Hypoglycemia: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S128-S145. doi:10.2337/dc25-S006

Symptoms and Signs of Hyperosmolar Hyperglycemic State

The primary symptom of hyperosmolar hyperglycemic state is altered consciousness, varying from confusion or disorientation to coma, usually as a result of extreme dehydration with or without prerenal azotemia, hyperglycemia, and hyperosmolality. In contrast to diabetic ketoacidosis, focal or generalized seizures and transient hemiplegia may occur.

Diagnosis of Hyperosmolar Hyperglycemic State

The diagnostic criteria for HHS are (1):

- Plasma glucose level $\geq$ 600 mg/dL (33.3 mmol/L)
- Serum osmolality > 300 mOsm/kg
- Absence of ketosis: Urine ketones < 2+ or beta-hydroxybutyrate < 3.0 mmol/L
- Absence of acidosis: pH $\geq$ 7.3 and serum bicarbonate $\geq$ 15 mEq/L (15 mmol/L)

Hyperosmolar hyperglycemic state is initially suspected when a markedly elevated glucose level is found in a fingerstick specimen obtained in the course of an evaluation of altered mental status. If measurements have not already been obtained, urine should be tested for ketones and the following should be measured in a blood sample:

- Serum electrolytes
- Blood urea nitrogen (BUN)
- Creatinine
- Glucose
- Ketones
- Plasma osmolality

Serum potassium levels are usually normal, but sodium may be low or high, depending on volume deficits.

Hyperglycemia may cause dilutional hyponatremia, so measured serum sodium is corrected by adding 1.6 mEq/L (1.6 mmol/L) for each 100 mg/dL (5.6 mmol/L) elevation of serum glucose over 100 mg/dL (5.6 mmol/L).

CLINICAL CALCULATORS

[Sodium Correction in Hyperglycemia \(Katz 1973\)](#)



BUN and serum creatinine levels are markedly increased.

Arterial pH is usually > 7.3, but occasionally mild [metabolic acidosis](#) develops due to lactate accumulation.

The fluid deficit can exceed 10 L, and acute circulatory collapse is a common cause of death. Widespread thrombosis is a frequent finding on autopsy and in some cases bleeding may occur as a consequence of [disseminated intravascular coagulation](#). Other complications include [aspiration pneumonia](#), [acute renal failure](#), and [acute respiratory distress syndrome](#).

It is not uncommon for features of both DKA and HHS to be present. For example, mild ketonemia often occurs in individuals with HHS. However, if significant ketonemia and acidosis are present in a patient with HHS/DKA overlap, [insulin](#) should be administered according to DKA protocols.

CLINICAL CALCULATORS

[Anion Gap](#)



Diagnosis reference

1. [American Diabetes Association Professional Practice Committee](#). 6. Glycemic Goals and Hypoglycemia: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S128-S145. doi:10.2337/dc25-S006

Treatment of Hyperosmolar Hyperglycemic State

- IV 0.9% saline
- Correction of any hypokalemia
- IV insulin (as long as serum potassium is ≥ 3.3 mEq/L [≥ 3.3 mmol/L])

All patients with hyperosmolar hyperglycemic state should be hospitalized and managed in the intensive care setting.

Treatment consists of IV saline, correction of hypokalemia, and IV insulin ([1](#), [2](#), [3](#)).

Treatment is 0.9% (isotonic) saline solution; 1000 mL is given in the first hour. Smaller boluses (500 mL) can be given if there is risk for exacerbation of heart failure or volume overload. Additional boluses may be needed for patients who are hypotensive.

After the first hour, intravenous fluids should be adjusted based on hemodynamic and electrolyte status but should generally be continued at a rate of 250 to 500 mL/hour.

A corrected sodium should be calculated. If the corrected sodium is < 135 mEq/L (< 135 mmol/L), then isotonic saline can be continued. If the corrected sodium is normal or elevated, then 0.45% saline (half normal) should be used.

[Dextrose](#) should be added once the glucose level reaches 250 to 300 mg/dL (13.9 to 16.7 mmol/L).

The rate of infusion of IV fluids should be adjusted depending on blood pressure, cardiac status, and the balance between fluid input and output.

[Insulin](#) is given as a 0.05 unit/kg/hour infusion after the first liter of saline has been infused and hypokalemia has been corrected. Hydration alone can sometimes precipitously decrease plasma glucose, so insulin dose may need to be reduced. A too-quick reduction in osmolality can lead to cerebral edema. Occasional patients with insulin-resistant type 2 diabetes with hyperosmolar hyperglycemic state require larger insulin doses. Once plasma glucose reaches 300 mg/dL (16.7 mmol/L), insulin infusion should be reduced to basal levels (1 to 2 units/hour) until rehydration is complete and the patient is able to eat.

If significant ketonemia and acidosis are present in a patient with HHS/DKA overlap, [insulin](#) should be administered according to DKA protocols.

Target plasma glucose is between 250 and 300 mg/dL (13.9 to 16.7 mmol/L). After recovery from the acute episode, patients are usually switched to adjusted doses of subcutaneous [insulin](#).

Potassium replacement is similar to that in [diabetic ketoacidosis](#): 40 mEq/hour for serum potassium < 3.3 mEq/L (< 3.3 mmol/L); 20 to 30 mEq/hour for serum potassium between 3.3 and 4.9 mEq/L (3.3 and 4.9 mmol/L); and none for serum potassium ≥ 5 mEq/L (≥ 5 mmol/L).

Treatment references

1. [American Diabetes Association Professional Practice Committee](#). 16. Diabetes Care in the Hospital: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S321-S334. doi:10.2337/dc25-S016
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3. [French EK, Donihi AC, Korytkowski MT](#): Diabetic ketoacidosis and hyperosmolar hyperglycemic syndrome: review of acute decompensated diabetes in adult patients. *BMJ*. 365:l1114, 2019. doi: 10.1136/bmj.l1114

Key Points

- Hyperosmolar hyperglycemic state (HHS) is a metabolic complication of diabetes mellitus characterized by severe hyperglycemia, extreme dehydration, hyperosmolar plasma, and altered consciousness.
- HHS can occur if infections, nonadherence, and certain medications trigger marked glucose elevation, dehydration, and altered consciousness in patients with type 2 diabetes.
- Patients have adequate insulin present to prevent ketoacidosis.
- The fluid deficit can exceed 10 L; treatment is 0.9% saline solution IV plus insulin infusion.
- Target plasma glucose in acute treatment is between 250 and 300 mg/dL (13.9 to 16.7 mmol/L).
- Give potassium replacement depending on serum potassium levels.

Alcoholic Ketoacidosis

Alcoholic ketoacidosis is a metabolic complication of alcohol use and starvation characterized by hyperketonemia and anion gap metabolic acidosis without significant hyperglycemia. Alcoholic ketoacidosis causes nausea, vomiting, and abdominal pain. Diagnosis is by history and findings of ketoacidosis without hyperglycemia. Treatment is IV saline solution and dextrose infusion.

Alcoholic ketoacidosis is attributed to the combined effects of [alcohol](#) and [starvation](#) on glucose metabolism.

Pathophysiology of Alcoholic Ketoacidosis

Alcohol diminishes hepatic gluconeogenesis and leads to decreased insulin secretion, increased lipolysis, impaired shunting of fatty acids to mitochondria, fatty acid oxidation, and subsequent ketogenesis, causing an [elevated anion gap metabolic acidosis](#). Growth hormone, epinephrine, cortisol, and glucagon are all increased. Plasma glucose levels are usually low or normal, but mild hyperglycemia sometimes occurs.

Symptoms and Signs of Alcoholic Ketoacidosis

Typically, an alcohol binge leads to vomiting and the cessation of alcohol or food intake for ≥ 24 hours. During this period of starvation, vomiting continues and abdominal pain develops, leading the patient to seek medical attention. [Pancreatitis](#) may occur.

Diagnosis of Alcoholic Ketoacidosis

- Calculation of anion gap
- Exclusion of other disorders

Diagnosis requires a high index of suspicion; similar symptoms in a patient with [alcohol use disorder](#) may result from [acute pancreatitis](#), methanol or ethylene glycol poisoning (see table [Symptoms and Treatment of Specific Poisons](#)), or [diabetic ketoacidosis](#) (DKA). Often, blood alcohol levels are no longer elevated when patients present with alcoholic ketoacidosis.

In patients suspected of having alcoholic ketoacidosis, serum electrolytes (including magnesium), blood urea nitrogen (BUN) and creatinine, glucose, ketones, amylase, lipase, and plasma osmolality should be measured. Urine should be tested for ketones. Patients who appear significantly ill and those with positive ketones should have arterial blood gas and serum lactate measurements.

The absence of hyperglycemia makes diabetic ketoacidosis improbable. Patients with mild hyperglycemia may have underlying [diabetes mellitus](#), which may be recognized by elevated levels of glycosylated hemoglobin (HbA1C).

Typical laboratory findings include:

- [High anion gap metabolic acidosis](#)
- Ketonemia and ketonuria
- [Hypokalemia](#)
- [Hypomagnesemia](#)

- [Hypophosphatemia](#)

Detection of acidosis may be complicated by concurrent [metabolic alkalosis](#) due to vomiting, resulting in a relatively normal pH; the main clue is the elevated anion gap. If history does not exclude toxic alcohol ingestion as a cause of the elevated anion gap, serum methanol and ethylene glycol levels should be measured.

The presence of calcium oxalate crystals in the urine also suggests ethylene glycol poisoning.

Lactic acid levels are often elevated because of hypoperfusion and the altered balance of reduction and oxidation reactions in the liver.

CLINICAL CALCULATORS

[Anion Gap](#)



Treatment of Alcoholic Ketoacidosis

- IV [thiamine](#) and other vitamins plus magnesium
- IV 5% [dextrose](#) in 0.9% saline

An IV infusion of 0.9% saline solution is given, usually with 5% dextrose. In patients with hypoglycemia, higher concentrations of dextrose may be given initially. Initial IV fluids should contain added water-soluble vitamins and magnesium, with potassium and phosphate replacement as required. Patients are also given [thiamine](#) 100 to 200 mg IV (500 mg in patients at increased risk of Wernicke encephalopathy) to prevent development of [Wernicke encephalopathy](#) or [Korsakoff psychosis](#) ([1](#), [2](#)).

Ketoacidosis and gastrointestinal symptoms usually respond rapidly. Serum potassium level should be monitored closely; as with [diabetic ketoacidosis](#), total body potassium is often depleted and intracellular shift with correction of acidosis via fluid resuscitation can lower serum potassium ([1](#)). Use of insulin is appropriate only if there is any question of atypical diabetic ketoacidosis or if hyperglycemia > 300 mg/dL (> 16.7 mmol/L) develops.

Treatment references

1. [Knight-Dunn L, Gorchynski J](#). Alcohol-Related Metabolic Emergencies. *Emerg Med Clin North Am*. 2023;41(4):809-819. doi:10.1016/j.emc.2023.07.003
2. [Long B, Lentz S, Gottlieb M](#). Alcoholic Ketoacidosis: Etiologies, Evaluation, and Management. *J Emerg Med*. 2021;61(6):658-665. doi:10.1016/j.jemermed.2021.09.007

Key Points

- Alcoholic ketoacidosis is caused by the combined effects of alcohol and starvation on glucose metabolism; it is characterized by hyperketonemia and elevated anion gap metabolic acidosis without significant hyperglycemia.
- Measure serum and urine ketones and electrolytes and calculate a serum anion gap.
- Treat initially with IV [thiamine](#) to prevent Wernicke encephalopathy or Korsakoff psychosis, then follow up with IV [dextrose](#) in 0.9% saline.

Drug Information for the Topic



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